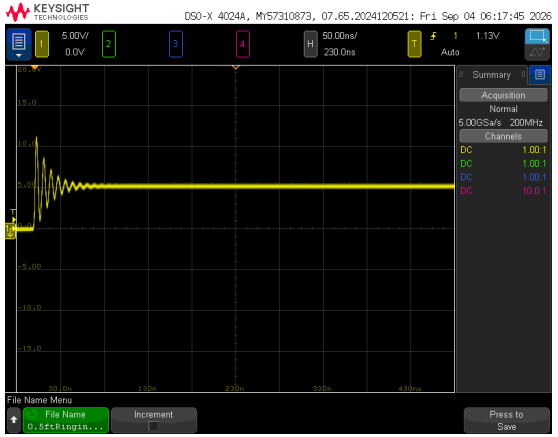
Lab 1.3

Hacking Ringing Reflections in Coax Cables

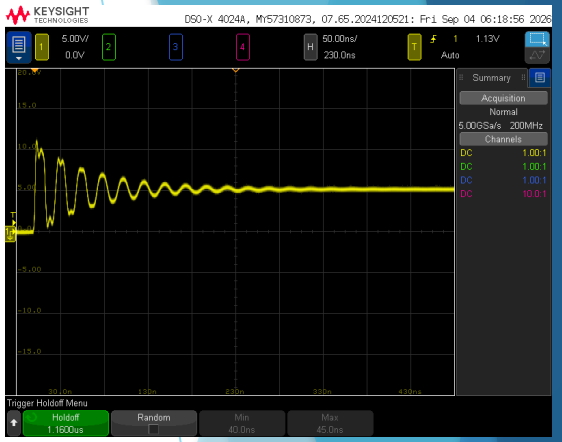
Leo Marcinkus and Jordan Bernhardt

Setup and Measurement

- Board 100, 3 Ω output, 5 V supply → RG58 coax → scope 1 MΩ input (no probe)
  - Two cable lengths, same cable type: 0.5 ft and 3 ft
- Scope: Keysight DSO 4024A, 5 GSa/s, 200 MHz
  - 50 ns/div for both captures so the two data sets share one time base
- **Trigger:** Rising edge at 1.13 V
  - 1.16 µs trigger holdoff for the 3 ft cable
  - 0.96 µs trigger holdoff for the 0.5 ft cable
  - Holdoff helps accomplish a lower threshold
  - Low threshold puts the start of the edge just before t = 0 (ADS only plots t ≥ 0)
- **Best practices applied:** Averaging off while setting up time base long enough to see the steady state (500 ns), 2000 points at 0.25 ns saved as CSV



0.5 ft cable, 50 ns/div



3 ft cable, 50 ns/div

What the Measurements Say

- **Same first peak, 11.1 V, and same steady state, 5.24 V**
- **Ring period:** 8.3 ns for 0.5 ft, 24.5 ns for 3 ft. Period ≈ 4 x TD
  - TD ≈ 2.1 ns and 6.1 ns
- Cable alone would give 0.77 ns and 4.6 ns (1.54 ns/ft). Both cables carry ≈ 1.3-1.5 ns extra: the SMA-BNC adapter, connectors, and the scope's input capacitance
  - 2.1 ns – 0.77 ns = 1.3 ns, 6.1 ns – 4.6 ns = 1.5 ns
- **Delay per foot from the difference:** (6.1 - 2.1) ns / (3 – 0.5) ft = 1.6 ns/ft
- **Decay:** peak-to-trough shrinks by 0.74–0.78 per period → ρ_{src} ≈ -0.87 → effective R_{src} ≈ 3-4 Ω, above the 1.7 Ω DC value from Lab 1.1. The lossless model must put the cable loss somewhere

Feature	0.5 ft	3 ft
1st peak / steady state	11.09 V / 5.24 V	11.11 V / 5.22 V
Ring period	8.3 ns	24.5 ns
Effective TD = period / 4	2.1 ns	6.1 ns
Cable-only TD (RG58 Spec: 1.54 ns/ft)	0.77 ns	4.6 ns
Extra delay (fixture + scope C)	1.3 ns	1.5 ns
R_{src} from decay	3.7 Ω	3.0 Ω

Bringing the Waveforms into ADS

- **Step 1 – header:** Replace the two scope header lines with BEGIN BLK and % time(1) v(1)
  - Keep time ≥ 0 (we shifted by +20 ns)
- **Step 2 – Dataset:** Tools → Data File Utility → Data File Tool, read as MDIF / Generic MDIF, write .ds
- **Step 3 – Source:** VtDataset with Expression = "v", output tied to ground through a dummy resistor so no node floats
- **Step 4 – Transient:** Start 0, Stop 500 ns, Min = Max step = 0.25 ns = the scope sample intervals
- **Rule #9:** The imported waveform must look exactly like the scope screen before any model is added

File	Rows	Step	Span	Header
0.5ftRingingHoldoff.txt	2000	0.25 ns	0–499.5 ns	v(1)
3ftRingingHoldoff.txt	2000	0.25 ns	0–499.5 ns	v(1)
0.5ft / 3ft Ringing1.3-2.txt	2000	0.5 ns	0–999.5 ns	v(1)
0.5ftRinging1.3.txt	2000	1 ns	0–1999 ns	v(1)
3ftRinging1.3.txt	2000	1 ns	0–1999 ns	voltage(1)

Questions?